

DODGE CITY, KS, USA

WMO: 724510

Lat: 37.7614N

Lon: 99.9686W

Elev: 2591

StdP: 13.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	9.8	-3.7	5.0	9.5	0.5	6.2	15.2	33.1	38.0	29.3	35.9	11.8	340	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	100.4	69.6	97.6	69.6	94.6	69.6	74.3	89.5	73.1	88.8	72.1	87.8	15.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	122.7	79.2	68.9	117.2	78.4	67.7	112.0	77.4	39.5	89.6	38.4	88.9	37.4	87.6	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.0	27.1	24.8	-2.9	104.6	4.4	2.7	-6.1	106.6	-8.7	108.2	-11.1	109.7	-14.3	111.6	
			DB												
			WB	-3.8	76.3	4.2	1.7	-6.8	77.5	-9.2	78.5	-11.6	79.5	-14.6	80.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	32.9	35.8	45.0	54.0	64.5	75.0	79.8	77.9	70.0	56.8	44.1	33.8
	DBStd	19.17	10.73	11.87	11.22	9.86	8.81	6.87	5.72	5.79	8.43	9.61	9.79	10.19
	HDD50	2023	535	412	225	68	5	0	0	0	2	43	227	506
	HDD65	4834	996	817	620	346	121	8	1	1	47	284	627	966
	CDD50	4185	4	15	72	190	453	750	925	865	602	255	50	4
	CDD65	1522	0	0	2	18	105	308	461	401	197	30	0	0
	CDH74	17952	1	13	80	353	1285	3570	5568	4454	2166	435	27	0
	CDH80	9191	0	3	12	114	537	1821	3152	2397	1007	145	3	0
Wind	WSAvg	12.7	12.0	12.7	13.8	14.6	13.3	13.2	11.7	11.0	12.5	12.7	12.4	12.0
Precipitation	PrecAvg	21.7	0.6	0.6	1.5	2.0	3.0	3.2	3.1	2.8	1.7	1.7	0.8	0.8
	PrecMax	32.4	2.6	2.8	8.8	8.1	10.3	9.1	9.1	7.4	6.8	6.5	3.7	4.3
	PrecMin	10.4	0.0	0.0	0.0	0.1	0.2	0.2	0.2	0.7	0.0	0.0	0.0	0.0
	PrecStd	4.6	0.6	0.6	1.5	1.7	1.8	1.9	2.0	1.8	1.3	1.5	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.8	75.0	81.3	88.9	96.8	103.2	104.0	102.8	98.9	90.0	77.7	66.5
		MCWB	47.8	50.9	54.4	58.8	64.4	66.6	69.9	69.4	67.2	63.5	53.8	48.7
	2%	DB	61.2	67.5	75.7	83.2	90.8	97.3	100.9	99.1	94.3	84.0	71.8	60.0
		MCWB	44.5	48.2	54.5	58.5	64.9	68.7	70.2	70.4	67.3	61.4	52.5	45.5
	5%	DB	55.0	60.9	69.6	77.7	86.1	93.7	98.2	96.0	90.1	78.8	66.3	54.6
		MCWB	41.8	45.8	51.6	57.0	64.0	69.3	70.4	70.3	67.8	59.5	50.4	42.8
	10%	DB	49.2	54.0	64.3	72.2	81.3	90.4	94.9	92.4	86.0	73.3	60.2	49.6
		MCWB	38.9	42.5	50.0	55.3	63.2	69.2	70.7	70.2	66.2	57.7	48.1	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	54.0	61.5	64.4	71.0	75.0	76.1	75.4	73.1	68.3	58.5	52.2
		MCDB	65.2	66.1	72.7	75.8	84.7	90.0	90.9	89.3	87.8	80.8	67.4	60.4
	2%	WB	45.3	50.1	58.0	61.7	68.6	73.2	74.5	73.7	71.2	65.5	55.3	46.9
		MCDB	59.3	65.1	67.2	73.9	82.2	88.8	89.8	88.9	85.9	74.6	64.8	57.5
	5%	WB	42.3	46.5	54.5	59.3	67.0	71.9	73.3	72.6	69.6	62.9	52.2	43.4
		MCDB	53.9	59.6	65.9	72.5	79.9	87.1	89.4	88.0	83.5	72.6	63.5	54.3
	10%	WB	39.5	43.1	50.9	56.8	65.2	70.5	72.2	71.4	68.1	60.1	49.3	40.3
		MCDB	48.6	54.1	63.8	70.5	78.0	85.1	88.5	87.0	81.3	70.7	61.1	48.9
Mean Daily Temperature Range	5% DB	MDBR	23.6	24.4	26.3	26.1	25.2	25.3	25.3	24.7	25.2	25.6	25.3	22.3
		MCDBR	32.6	35.5	34.5	34.5	31.4	29.6	29.7	29.7	29.3	32.4	32.9	30.3
		MCWBR	19.3	19.6	17.6	15.7	12.1	9.7	8.2	9.0	9.9	14.4	17.4	18.2
	5% WB	MCDBR	31.0	32.5	29.5	28.6	26.4	25.6	25.3	25.2	25.3	25.8	29.1	28.9
		MCWBR	18.9	18.6	16.5	15.3	12.4	10.3	9.1	9.3	10.1	13.5	16.4	17.9
Clear-Sky Solar Irradiance	taub		0.264	0.273	0.306	0.333	0.352	0.383	0.394	0.381	0.348	0.310	0.277	0.264
	taud		2.552	2.539	2.451	2.404	2.425	2.372	2.369	2.373	2.439	2.528	2.554	2.549
	Ebn at Noon		298	309	305	300	294	284	279	281	286	290	290	290
	Edh at Noon		23	27	33	36	36	38	38	37	33	27	23	22
All-Sky Solar Radiation	RadAvg		843	1102	1476	1818	2100	2309	2258	1987	1667	1227	920	744
	RadStd		72	118	121	102	133	131	95	107	99	142	85	60

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page